MIL-S-4938B(AL)
31 October 1983
SUPERSEDING
MIL-S-4938A(WC)
17 January 1972

MILITARY SPECIFICATION

SOLDERING IRON, ELECTRIC, 28-VOLT DC, 100-WATT

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers an electric soldering iron which operates from a nominal direct current (dc) power source of 28 volts, having a minimum rating of 100 watts, and is equipped with a 3/8 inch plug type tip.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specification, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-596/23 - Connector, Plug, Electrical, General Purpose, General Grade, 2 Pole, 2 Wire, 20 Amperes, 250 Volts, 50/60 RTZ

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3439

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MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U.S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29CFR 1910).

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

ANSI/NFPA No. 70 - National Electrical Code

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from Libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3 REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one or more, as specified, complete soldering iron(s) for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The soldering iron shall be new and of the manufacturer's latest design that will meet the performance requirements specified herein.

3.3 Materials. Materials shall be chosen to achieve their design objective as specified herein. All material shall be free of defects and imperfections that might affect the serviceability and appearance of the finished product. Materials not specifically designated shall be of good commercial quality, meeting all requirements specified herein. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The soldering iron may contain reclaimed materials provided such materials will not jeopardize its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use.

3.4 Construction. The soldering iron shall be complete so that it can be used for the operations specified herein.

* 3.4.1 Fastening devices. All screws, pins, bolts and similar fasteners used shall be installed with means for preventing loss of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

* 3.4.2 Threads. All threaded parts of the soldering iron shall conform to FED-STD-H28 and the applicable "Detailed Standards" section referenced therein.

* 3.4.3 Electrical connections. Electrical connections shall be mechanically secured and soldered, or brazed; except connections on screw type terminals which shall be mechanically secured. Rosin base fluxes shall be used in all soldering operations.

* 3.4.4 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.4.5 Interchangeability. All replaceable parts shall be manufactured to permit replacement or adjustment without the need for modification.

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3.4.6 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards which can be guarded without substantial interference to operation of the soldering iron. All guards shall be removable to provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the soldering iron shall comply with the standards promulgated under OSHA 2206 applicable to the soldering iron itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2 and 6.3).

3.5 Performance and product characteristics.

3.5.1 Vibration. The completely assembled soldering iron shall be capable of withstanding, without damage, a simple harmonic motion of 1/64 inch amplitude (total excursion of 1/32 inch), through a frequency range of 10 to 55 hertz in one continuous minute and from 55 to 10 hertz in the subsequent continuous minute. This procedure shall be continuously repeated for a period of one hour once in each of the three major axes. The soldering iron shall be capable of meeting the requirements of 3.5.2 through 3.5.7 after being subjected to this vibration.

* 3.5.2 Function. The soldering iron shall function under any of the following conditions or combination of conditions.

3.5.3 Wattage input. After an initial warm-up period of 20 minutes and within an applied voltage of 28 volts plus or minus 1 volt dc, the power input shall be not less than 100 watts nor more than 120 watts.

3.5.4 Heating time. Starting from a stabilized condition of 80 degrees Fahrenheit (F) plus or minus 5 degrees F, and operating on 28 volts plus or minus 1 volt, the soldering iron shall melt 1/8 inch diameter solid-core solder as follows:

Solder Composition	Maximum Time
50 percent tin, 50 percent lead	4 minutes

* 3.5.5 Melting efficiency. The soldering iron shall be capable of melting 1 ounce of solder consisting of 50 percent tin and 50 percent lead utilizing no more than 6,000 watt-seconds of energy at 28 volts plus or minus 1 volt.

Solder Composition	Energy Input
50 percent tin, 50 percent lead	6,000 watt-seconds

3.5.6 Continuous duty. During 5 hours of continuous operation at 28 volts plus or minus 1 volt, in an ambient air temperature of 70 degrees F plus or minus 5 degrees F, with the soldering iron at idle in a horizontal position, the wattage input shall vary no more than plus or minus 5 percent, the heating unit shall show no evidence of damage, and temperature rise on the surface of the soldering iron handle shall not exceed 55 degrees F.

3.5.6.1 Voltage range. The soldering iron shall be capable of operating at 24 volts dc for 1/2 hour and at 32.5 volts dc for 1/2 hour to cover the nominal voltage.

* 3.5.7 Electrical insulation breakdown. The soldering iron shall be insulated such that the iron is capable of withstanding a voltage of 500 volts for a period of ten seconds without the iron showing any visible deterioration. The soldering iron shall be capable of meeting the performance requirements specified herein after being subjected to the above.

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3.6 Details of components.

3.6.1 Heating unit. The heating unit shall be of the proper capacity, suitably insulated, and shall have a finish protected against corrosion.

3.6.2 Soldering tip. The soldering tip shall be a plug type, shall have a body diameter of 3/8 inch, shall have a pyramidal shape, and shall be fabricated of copper. The soldering tip shall be tinned to protect against oxidation of the copper. Means shall be provided to securely hold the tip in place.

3.6.3 Handle. The soldering iron handle shall be fabricated of either hardwood or a heat-resisting composition. The forward end of the handle shall be shaped to protect the operator's hand from contacting metal surfaces of the soldering iron. The other end of the handle shall have a beveled or flared opening to receive the power cord. The warning "FOR 28-VOLT DC ONLY" shall be permanently and conspicuously printed lengthwise along two opposite sides of the handle. The warning shall be printed in characters at least 3/8 inch in height.

3.6.4 Power cord. The power cord shall be type HPD, size No. 18 AWG, having a length of 6 feet plus or minus 4 inches, and conforming to the applicable requirements of NFPA No. 70. The connector plug shall conform with W-C-596/23 two-pole, 90 degrees, polarized. The power cord shall be provided with strain relief at the connection to the soldering iron.

3.7 Identification marking. The soldering iron shall be marked for identification in accordance with MIL-STD-130 and unless otherwise specified (see 6.2), shall include the National Stock Number.

3.8 Workmanship. Standards of workmanship shall be such that equipment specified herein shall comply with the requirements in section 3. All components, subassemblies, parts, and the final assembly shall comply with tests in section 4.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. First article inspection applies to the preproduction model or initial production item noted in 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination of one soldering iron per examination check list (4.4.2) and all tests under 4.5. Unless otherwise specified in the contract, tests under 4.5 shall be performed in order of sequence shown (4.5.1 through 4.5.6). Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

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4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of a. through c. as follows, and failure of any unit to pass an examination, test, or inspection shall be cause for rejection:

a. Examination (see 4.4.2).
b. Heating time test (see 4.5.4).
c. Packaging inspection (see 4.6).

* 4.4.1 Lot. All soldering irons manufactured under the same conditions and offered for delivery at one time shall be considered a lot for purposes of examination and testing.

* 4.4.1.1 Sampling for examination and tests. A random sample shall be selected from each lot offered to the Government in accordance with Inspection Level II of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 2.5 percent defective.

* 4.4.2 Examination. The soldering iron shall be examined to determine compliance with Section 3 herein and the contract, using the following check list as a guide:

EXAMINATION CHECK LIST

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Design	3.2
Materials	3.3
Reclaimed materials	3.3.1
Construction	3.4
Fastening devices	3.4.1
Threads	3.4.2
Electrical connections	3.4.3
Welding, brazing, or soldering	3.4.4
Interchangeability	3.4.5
Safety and health requirements	3.4.6
Heating unit	3.6.1
Soldering tip	3.6.2
Handle	3.6.3
Power cord	3.6.4
Identification marking	3.7
Workmanship	3.8

NOTE: A certificate of conformance may be provided to satisfy the examination requirement where more than a visual or dimensional verification is necessary.

* 4.5 Performance tests. The soldering iron shall be functionally tested for the following characteristics (see 3.5).

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4.5.1 Vibration test. The soldering iron shall be rigidly mounted on a vibration machine and subjected to the vibration conditions in 3.5.1. The soldering iron shall then be removed from the vibration machine, inspected for evidence of damage, and tested in accordance with 4.5.2 through 4.5.6.

* 4.5.2 Electrical insulation breakdown test. The soldering iron shall have 500 volts dc applied for a period of ten seconds at ambient temperature between the current carrying parts and all external metal parts of the iron to determine compliance with 3.5.7.

* 4.5.3 Voltage range test. The soldering iron shall be placed in a horizontal position and allowed to idle on 24 volts dc for 1/2 hour, followed with 32.5 volts dc for 1/2 hour, thus covering the nominal voltage in accordance with 3.5.6.1. Wattage in accordance with 3.5.3 shall be checked periodically. Temperature on the exterior of the soldering iron shall be determined prior to as well as after the test. The soldering iron shall then be tested in accordance with 4.5.4.

4.5.4 Heating time test. The soldering iron, starting from a stabilized condition of 80 degrees F plus or minus 5 degrees F, shall be operated to determine compliance with 3.5.4.

4.5.5 Melting efficiency test. The soldering iron shall be attached to a framework at an angle of 30 degrees from a horizontal plane, point downward. Solder shall rest in a slide which has been built into the framework to position the longitudinal axis of the solder at an angle of 90 degrees from the longitudinal axis of the soldering iron (see Figure 1). A 15-inch length of 1/8 inch diameter, solid-core, general purpose 50-50 tin-lead solder shall be used. Testing shall be conducted as follows. The soldering iron, operating on 28 volts plus or minus 1 volt, shall be allowed to preheat for a period of 5 minutes. Then the solder shall be placed in its slide and allowed to feed to the soldering iron tip by gravity. When 50 percent of the solder has been melted, the soldering iron shall be turned off. The solder shall be allowed to continue melting so that the reserve heat of the soldering iron is utilized. The melting efficiency shall comply with the requirements of 3.5.5.

4.5.6 Continuous duty test. The soldering iron shall be placed in a horizontal position and allowed to idle on 28 volts plus or minus 1 volt dc for a continuous period of 5 hours, while in an ambient air temperature of 70 degrees F plus or minus 5 degrees F for compliance with 3.5.6. Wattage input shall be checked periodically. The temperature on the exterior of the soldering iron handle shall have been determined prior to this test, and shall again be determined at the conclusion of this test. Temperature rise, at the hottest point on the handle surface, shall not exceed 55 degrees F. Wattage input shall vary no more than plus or minus 5 percent. The soldering iron shall at no time exceed 120 watts for compliance with 3.5.3.

* 4.6 Packaging inspection. An examination of the packaging shall be performed to determine compliance with section 5. The lot shall consist of packages or shipping containers as applicable. The sample shall be one package or one shipping container. The inspection level shall be Level S-2 and the Acceptable Quality Level (AQL) shall not exceed 4.0, expressed percent defective.

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5 PACKAGING

5.1 Preservation. Preservation shall be in accordance with MIL-W-45562. The level of preservation shall be as specified (see 6.2).

5.2 Packing. Packing shall be in accordance with MIL-W-45562. The level of packing shall be as specified (see 6.2).

5.3 Marking. Marking shall be in accordance with MIL-W-45562.

6 NOTES

6.1 Intended use. The soldering iron covered by this specification is intended for light soldering work in locations where only a nominal voltage of 28 volts dc is available.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. First article, when required (see 3.1).
c. Special safety devices, when required (see 3.4.6 and 6.3).
d. Identification marking, if different (see 3.7).
e. First article inspection, if different (see 4.3).
f. Quality conformance inspection (see 4.4).
g. Selection of applicable level of preservation and packing required (see 5.1 and 5.2).

* 6.3 Safety and health determination. In order that equipment integrated into the user's operational environment will comply with OSHA limitation and control of noise levels, radiation, electromagnetic emmision, noxious vapors, heat, etc., as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified by the user/requisitioner (see 6.2).

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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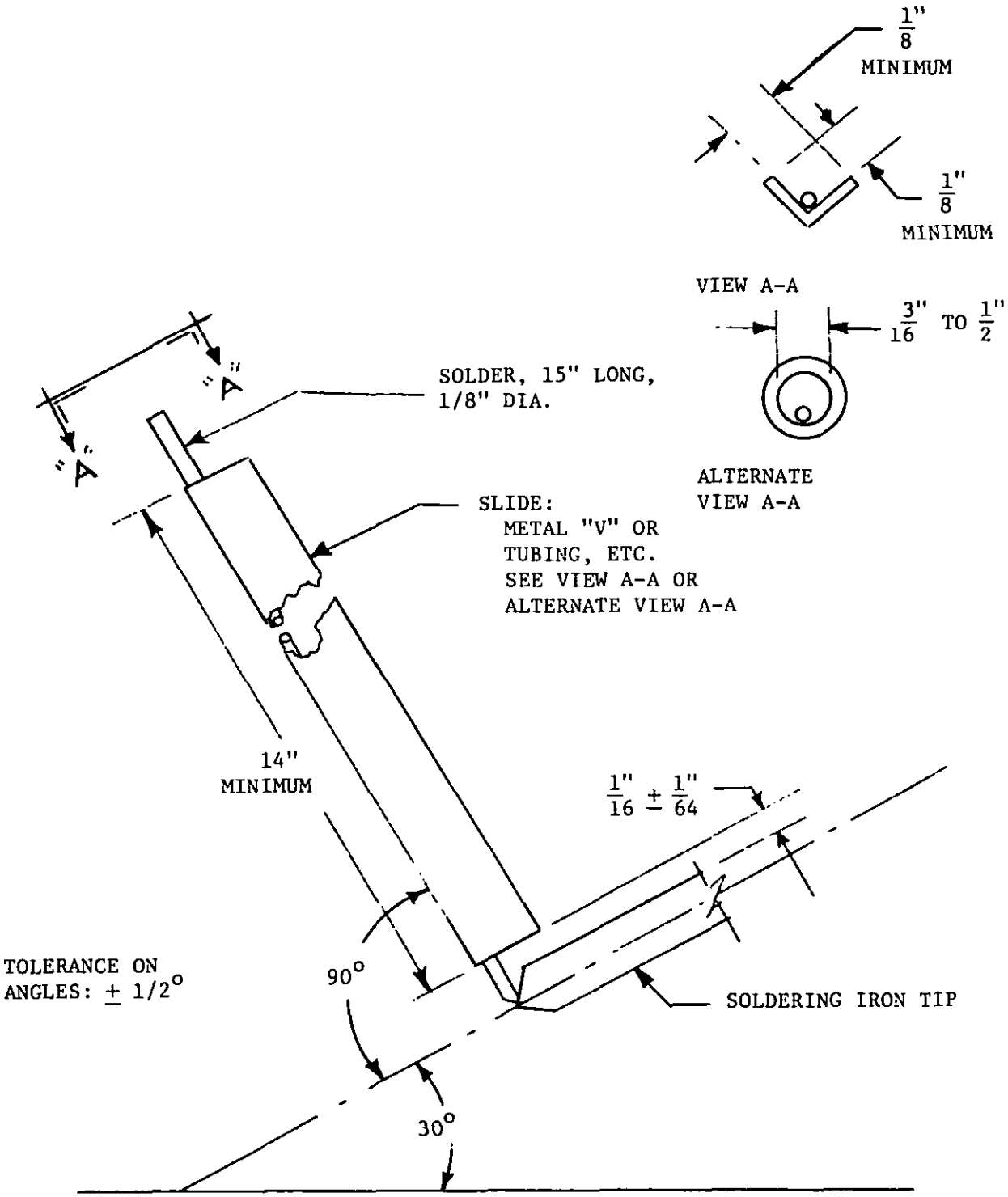


Figure 1. Melting efficiency test.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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